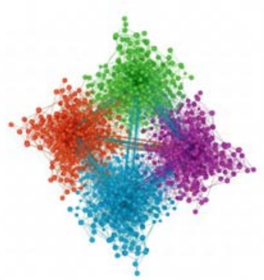
A4.3.4 Unsupervised Learning: Clustering



A4.3.4 Unsupervised Learning: Clustering

A4.3.4 details unsupervised learning's clustering techniques, which group data points based on inherent feature similarities. A practical application is segmenting a customer base using purchasing data.

Introduction

So far, we have focused on Supervised Learning, where our data has labels and we are trying to predict a known outcome. But what if we have a large dataset with no labels at all? How can we find meaningful patterns? This is the domain of **Unsupervised Learning**, and its most common task is **Clustering**.

What is Clustering?

Clustering is an unsupervised learning technique that involves grouping a set of data points in such a way that points in the same group (called a **cluster**) are more similar to each other than to those in other clusters.

The algorithm doesn't know what the "correct" groups are beforehand. Its job is to look at the features of the data and discover these natural groupings on its own.

- **Analogy:** Imagine being given a huge, mixed box of fruit and being told to "organise it." Without any specific instructions, you would likely start making piles based on similarity: a pile of apples, a pile of bananas, a pile of oranges. You have performed clustering.

How Clustering Works: The k-means Algorithm

One of the most popular and intuitive clustering algorithms is **k-means**. Its goal is to partition the data into 'k' distinct clusters, where 'k' is a number you choose beforehand.

Here's the conceptual process:

1. **Choose 'k':** You decide how many clusters you want to find (e.g., k=3).
2. **Initialize Centroids:** The algorithm randomly places 'k' points on the graph. These are the initial centers of your clusters, called **centroids**.
3. **Assign Points:** Each data point in your dataset is assigned to the closest centroid. This creates 'k' initial clusters.
4. **Update Centroids:** The algorithm calculates the new cenre of each cluster by finding the average position of all the points within it. It moves the centroid to this new average position.
5. **Repeat:** Steps 3 and 4 are repeated until the centroids no longer move. At this point, the clusters are stable, and the algorithm has finished.

Real-World Application: Customer Segmentation

Clustering is a cornerstone of marketing and business analytics. Companies use it to segment their customer base and understand their users better.

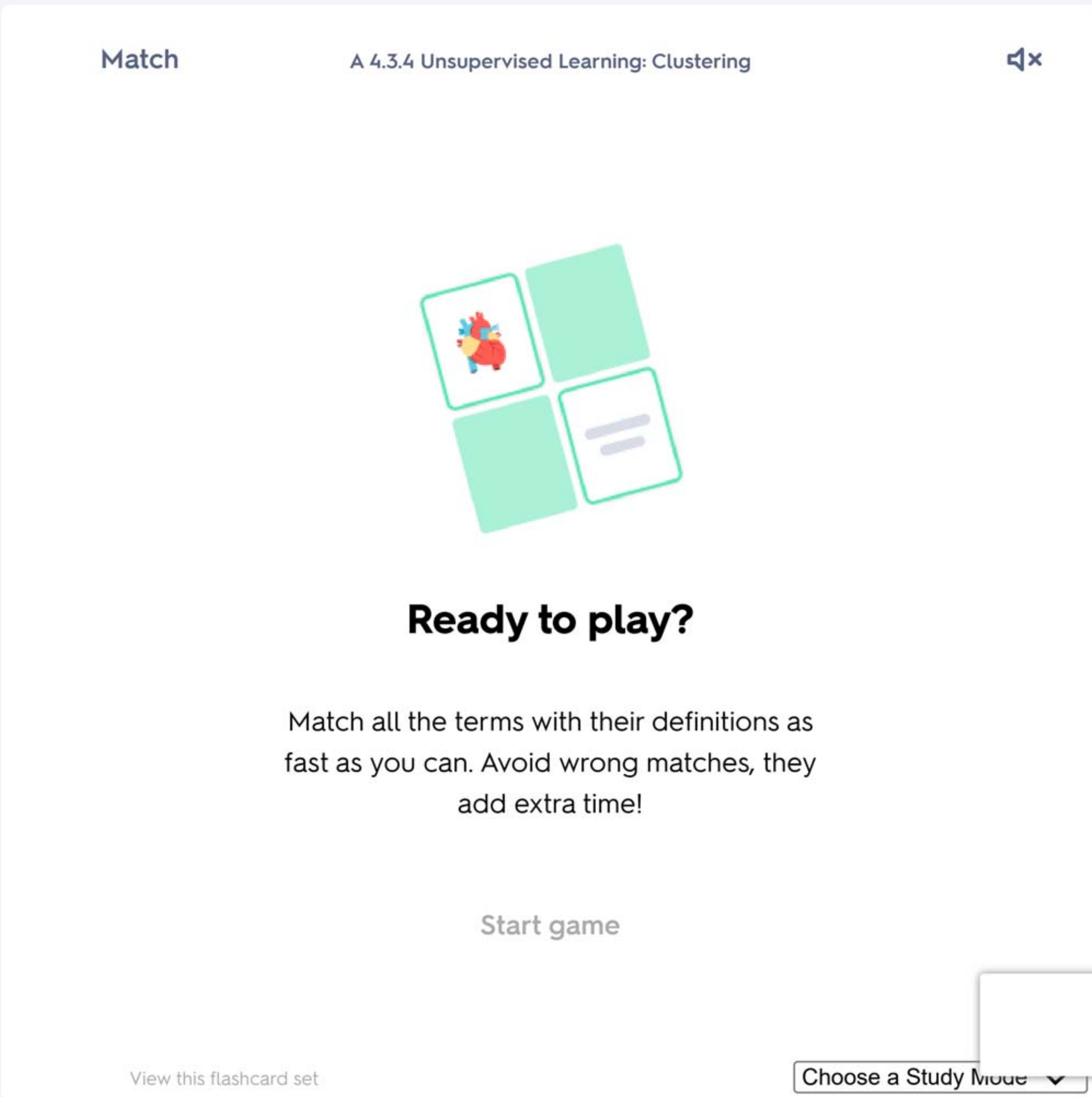
- **Scenario:** An e-commerce website has purchasing data for thousands of customers but doesn't know what "types" of customers it has.
- **Clustering in Action:** They can use a clustering algorithm like k-means on data features like `average_spend`, `frequency_of_purchase`, and `types_of_products_bought`.
- **The Result:** The algorithm might discover several distinct clusters, which the company can then analyse and label:
    - **Cluster 1: "High-Value Spenders":** Customers who buy expensive items frequently.
    - **Cluster 2: "Bargain Hunters":** Customers who mostly buy items that are on sale.

- **Cluster 3: "One-Time Buyers":** Customers who made one purchase and never returned.

This allows the company to create targeted marketing campaigns. They can send exclusive offers to the "High-Value Spenders" and "come back" discounts to the "One-Time Buyers."

Flashcards

Use this Quizlet to learn all the key terms in this section.



Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

Clustering is an example of which main type of machine learning?

A. Unsupervised Learning

B. Supervised Learning

C. Transfer Learning

D. Reinforcement Learning

2

What is the primary goal of a clustering algorithm?

A. To group similar data points together based on their features.<

B. To classify data based on pre-existing labels.

C. To predict a continuous numerical value.

D. To learn a policy through trial and error.

3

In the k-means algorithm, what does the 'k' represent?

A. The number of times the algorithm is run.

B. The number of clusters to be found

C. The number of features for each data point.

D. The number of data points in the dataset.

4

The center of a cluster in the k-means algorithm is called a:

A. Neighbour

B. Node

C. Centroid

D. Leaf

5

The k-means algorithm assigns each data point to a cluster based on:

A. A set of if-then rules.

B. A majority vote of its neighbours.

C. A random assignment.

D. Its distance to the nearest centroid.

6

Which of the following is a key real-world application of clustering?

A. Segmenting customers into different groups for marketing.

B. Training a robot to navigate a maze.

C. Predicting stock prices.

D. Classifying images of cats and dogs.

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The initial placement of centroids in the k-means algorithm is typically:

A. On top of the first 'k' data points.

B. In the corner of the graph.

C. Random.

D. Based on a pre-defined rule.

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When does the iterative process of the k-means algorithm stop?

A. When the centroids no longer move their position between steps.

B. When all data points have been assigned to a cluster.

C. After a fixed number of steps, usually 100.

D. When the user manually stops it.

9

The data used in a clustering task is:

A. Always text-based.

B. Labelled with the correct categories.

C. Unlabelled

D. A continuous stream of rewards.

10

A streaming service uses an algorithm to group movies into collections like "Quirky Comedies" and "Dark Sci-Fi Thrillers" without any pre-existing genre tags. This is an example of:

A. Reinforcement Learning

B. Classification

C. Regression

D. Clustering

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